

9(a)(i)	increase flux linkage (with secondary coil) / to reduce flux loss	B1
9(a)(ii)	e.m.f. (induced only) when flux (in core/coil) is changing	B1
	constant / direct voltage gives constant flux / field	B1
9(b)(i)	$N_S / N_P = V_S / V_P$	C1
	$N_S = (52 / 150) \times 1200$ = 416 turns	A1
9(b)(ii)	0 ms or 7.5 ms or 15.0 ms or 22.5 ms	A1
9(c)(i)	<i>either</i>	
	mean power = $V^2 / 2R$ and $V = 52$ (V)	C1
	$R = 52^2 / (2 \times 1.2)$ = 1100 (1127) Ω	A1
	<i>or</i>	
	mean power = V^2 / R and $V = 52 / \sqrt{2}$ (= 36.8 V)	(C1)
	$R = 36.8^2 / 1.2$ = 1100 Ω	(A1)
9(c)(ii)	sinusoidal shape with troughs at zero power	B1
	only 3 'cycles'	B1
	each 'cycle' is 2.4 W high and zero power at correct times	B1